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Barriers to Pre-Antibiotic Culture Practice: A Mixed-Methods Study Among Physicians at Sher-e-Bangla Medical College, Barishal

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Research question: What are the barriers to pre-antibiotic culture testing among the physicians of the Department of Medicine at Sher-e-Bangla Medical College (SBMC), Barishal?

Introduction

Antibiotics are among one of the most essential tools in modern medicine, but their effectiveness is being compromised by antimicrobial resistance (AMR). AMR is a major global health threat that develops due to inappropriate use of antibiotics[1]. This can be avoided by obtaining cultures before starting antibiotics. Because it identifies the causative organism and its susceptibility, guiding subsequent appropriate therapy [2]. However, this step is often missed by physicians. A study from Pune, India shows that only 37.5% of the general practitioners ordered culture and sensitivity before prescribing antibiotics. They reported time constraints, cost, uncertain patient follow-up, and poor lab access as barriers [3]. Another study in Korea shows that in Korean long-term care hospitals, only 16.0% always tested before starting antibiotics [4]. A German study involving nationwide physicians depicted that though physicians perceived blood culture as an important diagnostic, yet not ordered due to lack of time, low priority and limited laboratory access [5]; a Nigerian study found similar barriers [6]. Evidence from Bangladesh remains limited. A point prevalent survey involving four Bangladeshi government hospitals depicted the context of local concern — about 78% of the inpatients were on empirical antibiotic therapy without any preceding culture testing [7]. Another national study found microbiology results were not routinely used to guide antibiotic prescribing [8], but how often cultures are actually ordered locally, and why they are skipped, remains unclear. So, this study aims to identify these barriers among the physicians of the Department of Medicine at Sher-e-Bangla Medical College, Barisal, and to compare their reported KAP with documented culture ordering.

Literature Review

AMR can develop through multiple interacting mechanisms such as mutation, horizontal gene transfer, and selection pressure from antibiotic exposure [9], [10]. Long-term use of broad-spectrum antibiotics without culture-guided de-escalation may select for resistant organisms [9], [10]. Stewardship guidance recommends obtaining cultures before antibiotics when clinically indicated [2], yet evidence across settings shows this is inconsistently followed. In a study conducted in Pune, India, good knowledge did not translate into practice: only 37.5% tested before prescribing [3]. In Korea, only 16.0% of physicians in long-term care hospitals always tested first [4]. Some blood-culture-specific studies also show the same pattern — German inpatient physicians valued blood cultures but often skipped them, citing time and lab-access barriers [5], and Nigerian physicians showed similar gaps in a Nigerian study [6]. These barriers imply that knowledge alone doesn't contribute to the gap. Patient and system-level factors matter too: U.S. primary-care physicians cited patient pressure and access barriers when skipping UTI cultures [11]. Locally, a point prevalence survey in Bangladeshi government hospitals found that broad-spectrum Watch-group antibiotics made up most use, with third-generation cephalosporins the most common class, and its authors linked this pattern to empirical prescribing with limited test-directed therapy [7]. In another study of Bangladesh, microbiology results were found to be underused in prescribing decisions, with cost also influencing treatment [8]. How reported and documented practice compare locally, and why any

gap exists, remains unstudied — this study addresses that using a physician knowledge, attitude, and practice (KAP) survey, a chart audit, checklist and open-ended question on barriers.

Research Objectives

Primary objective: To identify the barriers to pre-antibiotic culture testing among the physicians of the Department of Medicine at Sher-e-Bangla Medical College, Barishal.

Secondary objectives:

1. To assess physicians' knowledge, attitude, and self-reported practice regarding pre-antibiotic culture testing, using knowledge, attitude, and practice (KAP) scores.
2. To determine the documented rate of pre-antibiotic culture ordering through retrospective review of admission records.
3. To compare reported practice with the documented ordering rate to quantify the know-do gap.
4. To identify and categorize the barriers physicians report for not ordering cultures before antibiotics.

Research Methodology

Study Design

Mixed-methods, convergent design.

Quantitative data will be collected from the physician survey (KAP scores and a structured barrier checklist) and a retrospective chart audit. Qualitative data will come from one open-ended survey question asking physicians to describe barriers in their own words. Both strands will be collected during the same study period and integrated during interpretation.

Setting and Study Population

The study will be conducted in the Medicine Department of Sher-e-Bangla Medical College, Barisal. The survey population consists of 143 physicians involved in inpatient admission and antibiotic prescribing in the Medicine Department.

Quantitative Component 1: Physician Survey

Total enumeration will be used — all 143 eligible physicians will be approached, so no sampling is needed. Sample size determined by following formula — $n_0 = Z^2p(1-p)/d^2$ ($Z=1.96$, $p=0.50$, $d=0.05$) gives $n_0 = 384$; adjusted for the finite population ($N=143$), $n \approx 104$; allowing 10% non-response, ~116 responses are needed — covered by approaching all 143.

A semi-structured Google Forms questionnaire will include demographics (designation and years of experience), knowledge items, attitude items (5-point Likert scale), clinical-vignette practice items, a barrier checklist, and one open-ended question. Knowledge and practice items are adapted from published KAP tools [3], [5]; checklist items are drawn from barriers reported in the literature [3], [5], [11] (time, cost, lab access, low priority, patient expectations). Vignettes describe a patient where a culture is indicated; only "order a culture before starting antibiotics"

counts as guideline-concordant [2]. The questionnaire will be reviewed with the study mentor and pilot-tested on physicians outside the study population before use.

Quantitative Component 2: Chart Audit

A retrospective review of adult Medicine admission charts with a documented antibiotic prescription, from the preceding six months, selected by convenience sampling until the target is reached. Using the same formula, sample size, $n_0 = 384$; allowing 10% for incomplete records, the target becomes ~427 charts. A structured extraction form will be used to record patient age and diagnosis, antibiotic prescribed, whether a culture was documented, culture type (blood, urine, sputum, wound, other), timing relative to the first antibiotic dose, and remarks.

Qualitative Component: Open-Ended Question

Qualitative data come from the questionnaire's open-ended question, capturing barriers the checklist may miss. All physicians who answer it will be included; no interviews or focus groups are planned.

Inclusion criteria: Inpatient physicians in Medicine involved in admission and prescribing will be included in the survey; adult Medicine admissions from the preceding six months with a documented antibiotic prescription will be included in the chart audit.

Exclusion criteria: Physicians who do not consent or are unavailable during data collection will be excluded from the survey; incomplete records and cases where antibiotics were already started before SBMC admission will be excluded from the chart audit.

Data Analysis

Questionnaire responses will be collected through Google Forms, compiled in Google Sheets, and exported to Jamovi (free, R-based statistical software) for analysis.

Quantitative Analysis

23 Results will be considered statistically significant when the p-value is below 0.05. Knowledge scores are the percentage of correct answers with a score of ≥55.6% considered good knowledge, while a score <55.6% considered poor knowledge, based on the cutoff adapted from Rajopadhye et al. [3]; attitude uses Likert averages; practice is the percentage of guideline-concordant
30 vignette answers. Chi-square/Fisher's exact test will check associations with physician characteristics; Spearman's correlation will be used to assess the associations between
15 knowledge, attitude, and practice scores. Mann-Whitney U and Kruskal-Wallis tests will compare scores across groups. Barrier checklist responses will be reported as frequencies and percentages and ranked from most to least commonly reported, then fed into the joint display. The know-do gap is calculated by comparing the % guideline-concordant survey answers with the % of charts showing a culture ordered before antibiotics, reported with a 95% confidence interval.

Qualitative Analysis

Open-ended responses will be analyzed thematically following Braun and Clarke [12] — coding responses, grouping into themes, and checking themes against the data, with a second team member reviewing a subset of codes.

Triangulation

A joint display will bring the barrier checklist, open-ended responses, KAP survey, and chart audit together by barrier/theme, showing where self-reported and documented practice converge or diverge, so self-report is not simply assumed to reflect actual practice.

Ethical Considerations

Ethical approval will be sought from the Institutional Review Board of Sher-E-Bangla Medical College , with hospital authority permission for the chart review. Physicians will give informed consent electronically before the survey; participation is voluntary and anonymous, with no physician identifiers collected, and withdrawal after submission is not possible since responses are anonymous — this will be stated during consent. A waiver of patient consent will be sought for the chart review, since it involves no patient contact and uses anonymized data with study-specific case IDs. The main anticipated risk is professional discomfort discussing prescribing practices; findings will be reported only in aggregate.

Project Practicalities

Timeline

The study period will take 49 weeks; each activity timeline is determined as follows — proposal finalization (Weeks 1–2), ethics and hospital approval (Weeks 3–6), instrument finalization (Weeks 7–8), pilot testing (Weeks 9–10), revision and team training (Weeks 11–12), data collection (Weeks 13–38), data cleaning (Weeks 39–40), analysis (Weeks 41–43), and report/manuscript writing (Weeks 44–49).

Gantt chart:

Activity	Wk 1–2	Wk 3–6	Wk 7–8	Wk 9–10	Wk 11–12	Wk 13–38	Wk 39–40	Wk 41–43	Wk 44–49
Finalize & submit proposal	■								
Ethics & hospital approval		■							
Finalize instruments & sample size			■						
Pilot-test instruments				■					
Revise instruments; train team					■				
Data collection (survey + chart audit)						■			
Data cleaning & de-identification							■		
Statistical & qualitative analysis								■	

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Activity	Wk 1–2	Wk 3–6	Wk 7–8	Wk 9–10	Wk 11–12	Wk 13–38	Wk 39–40	Wk 41–43	Wk 44–49
Report / manuscript writing									

Figure 1. Study timeline; Shaded cells indicate the weeks during which each activity takes place.

Budget

Item	Cost (BDT)	Cost (USD, approx.)
Mobile data / internet (data collectors, 6-month study period)	3,600	~$29.03
Contingency (10%)	360	~$2.90
Total	**3,960**	**~$31.94**

USD conversion is approximate, based on 1 USD ≈ 124 BDT for budgeting purposes.

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Data Privacy and Security: study-specific case IDs will be used for chart data; survey responses will be stored in a password-protected Google account accessible only to the research team; then analysis will run on a de-identified copy; data will be retained and disposed of per institutional policy.

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Roadblocks and Potential Limitations: Non-response and convenience sampling may introduce bias, and single-department findings may not apply in other departments/ institutions. Self-reported practice items may be affected by recall or social-desirability bias, and an undocumented culture does not always mean none was sent, which could overstate the gap. Since charts do not identify the prescribing physician, the survey and chart audit can only be compared at the group level, not matched individually, and the qualitative component is limited to one open-ended question rather than interviews. The investigator's prior observations may also influence data extraction, which will be mitigated using predefined criteria and independent review of a subset of records.

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Expected Outcomes: This study is expected to provide the documented rate of pre-antibiotic culture ordering among Medicine admissions, a profile of physicians' knowledge, attitudes, and self-reported practice, a comparison of reported versus documented practice showing where they converge or diverge, and a categorized description of the barriers physicians report for not ordering cultures before antibiotics.

Post-Program Plan: Findings will be shared with the Medicine Department and hospital authority to inform local stewardship discussions. The prepared abstract will target well-known conferences; after writing the full manuscript, aim is to publish it in a peer-reviewed journal in infectious diseases or antimicrobial stewardship.

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